A talking Pakistani banknote reader for blind people, tested beyond its own dataset

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Live demo: https://boltanote.vercel.app · Code, weights & photos: https://github.com/Hassan-Ali-Mehdi-3024/boltanote · Video: https://youtu.be/Je7tUmXsgO8

Abstract. Blind people in Pakistan (about 1.1 million [1]) cannot tell banknotes apart without help, and a Rs 500 / Rs 5,000 mix-up costs Rs 4,500. The reader is a web app that reads the phone camera about four times a second and says the note's value aloud in Urdu (پانچ سو روپے) or English, offline once loaded. We fine-tune MobileNetV3-Large [2] and MobileNetV4-Conv-Medium [3] and train a CNN from scratch on a public dataset of 3,611 phone photos (15 classes: seven notes × front/back, plus "no note"). Before measuring anything we checked how much the public data repeats itself: 339 of the 402 test photos in a second public currency dataset are near-copies of the first, leaving 63 genuinely new photos. The deployed MobileNetV3-Large scores 100.0% on unseen photos from the training dataset but 92.1% on photos by other people. Because saying a wrong amount is worse than saying nothing, the app speaks only above a confidence threshold calibrated on a stress-tested validation set.

1. Problem

Cash is still how most everyday purchases in Pakistan are paid. The seven notes in circulation (Rs 10 to Rs 5,000) differ only modestly in size, so a blind person usually has to ask someone and trust the answer. Commercial apps exist for some currencies, but an assistant for Pakistani notes should work on cheap phones, speak Urdu, and need no installation. The cost of errors is asymmetric: silence just means "move the note and try again", while a wrong value can cost real money.

Contributions. (1) A working Urdu-speaking reader that runs in any phone browser. (2) A measurement of hidden duplication in public currency datasets, and an evaluation that goes beyond the training dataset: other photographers and our own photos. (3) A speaking rule that trades coverage for safety, calibrated without touching test data. (4) An accuracy-versus-speed comparison of three architectures for phones.

2. Data

Training data. The Pakistan Image Currency Dataset [7] (CC0) has 3,611 phone photos of the 2005–2008 note series, each value front and back, many folded, partly visible, or in a hand, plus 280 "other" photos. We add 1,000 Imagenette photos [6] (100 per class) to the "no note" class so the model learns to stay silent.

Duplicate check. We computed a 64-bit perceptual hash [9] of every photo and joined photos within 8 bits of each other. Inside the training dataset, near-copies are kept on the same side of the 70 / 15 / 15 per-class split (seed 42). Against the Roboflow currency dataset [8], which we

intended as an independent test, **339 of 402** single-note test photos turned out to be near-copies of training photos; only 63 remain as a genuine external test. Without this check that external test would have reported near-perfect accuracy.

Our own photos. Being collected: about 300 photos of our own notes under real conditions, used only for testing.

Split	Photos
Train	3,429
Validation	735
Same-dataset test	727
External test (other photographers)	63
Our own photos	–

3. Method

Each model sees the whole camera frame squeezed to 224 × 224 (the same preprocessing in training, `predict.py` and the browser) and outputs 15 class probabilities; front and back are separate classes because they look nothing alike, and the app adds them per value. Augmentation is designed around how blind users actually hold notes: any rotation, too close or too far, half the note out of frame, fingers over it, dim or yellow light, motion blur. Mirror flips are excluded because the camera never sees a mirrored note.

Setting	Value
Input	RGB photo, whole frame squeezed to 224 × 224, values in [0, 1] (cached at 320 px for augmentation)
Classes	15 = 7 values × front/back + none
Optimiser	AdamW, weight decay 0.05, label smoothing 0.1
Learning rate	1e-3 (MobileNets), 2e-3 (scratch); 5% warm-up then cosine
Epochs	40 (MobileNets), 80 (scratch); best epoch by validation value accuracy
Batch	64, bfloat16 mixed precision, 1 × NVIDIA L4 (Colab)
Augmentation (GPU)	rotation ±180°, zoom 0.85–1.9×, stretch 0.8–1.25, shift ±18%, 0–2 finger/shadow patches, brightness/contrast ±35%, gamma ±30%, saturation ±40%, colour cast ±8%, noise, blur (p = 0.3); no mirror flips
Speaking threshold	lowest threshold with <= 0.2% (1 in 500) wrong-value rate on 4 augmented passes of the validation set

> **When to speak.** On ordinary validation photos the model is almost never wrong, so any threshold looks safe there (the lowest we searched, 0.40, already gives no more than 1 wrong answer in 500). Modern networks are over-confident on data like their training data [10]. We therefore calibrate on a *stress* validation set: each validation photo four times with the training augmentations. Because a wrong amount costs real money, we require at most 1 wrong value in 500 there (0.2%); the lowest threshold that achieves it is **0.68**. The app additionally requires the same answer in three consecutive frames before speaking.

4. Results

Model	Params	ONNX	CPU ms	Same-dataset test	Other photographers	Own photos	Threshold
CNN from scratch	2.36M	9.4 MB	88	99.6%	82.5%	–	0.76
MobileNetV3-Large **(deployed)**	4.22M	16.9 MB	15	100.0%	92.1%	–	0.68
MobileNetV4-Conv-Medium	8.45M	33.7 MB	37	99.9%	92.1%	–	0.70

All models are near-perfect on unseen photos from the training dataset, which mostly shows how similar those photos are to each other. The informative columns are the other two: on photos by other people the deployed model drops by 7.9 points, and the scratch CNN, which matched the pretrained models on the same-dataset test, falls to 82.5%. ImageNet pretraining buys robustness that the in-dataset test cannot show. MobileNetV3-Large runs in 15 ms per photo on a 2-core CPU, MobileNetV4-Conv-Medium in 37 ms (2.5× slower, 2.0× the parameters). The test on our own photos was still being collected when this version was written.

Test set	Photos	Value accuracy	Says wrong value	Speaks on note photos	Correct when speaking	Speaks with no note
Same-dataset test	727	100.0%	0.0%	99.4%	100.0%	0.0%
Other photographers	63	92.1%	0.0%	87.3%	100.0%	–

Value	Same-dataset test	Other photographers
Rs 10	100.0%	100.0%
Rs 20	100.0%	100.0%
Rs 50	100.0%	90.0%
Rs 100	100.0%	100.0%
Rs 500	100.0%	55.6%
Rs 1000	100.0%	100.0%
Rs 5000	100.0%	100.0%
no note	100.0%	–

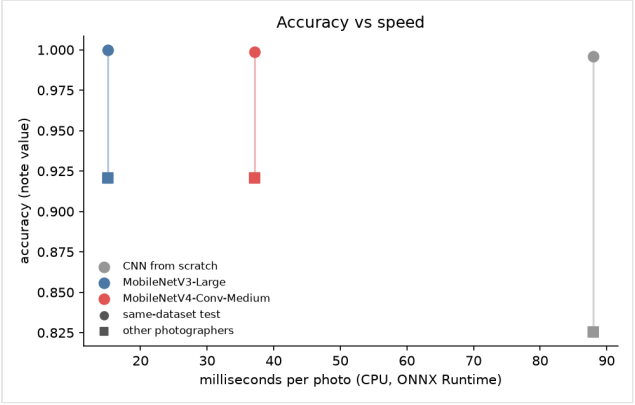


Figure 1. Accuracy on each test set against CPU time per photo.

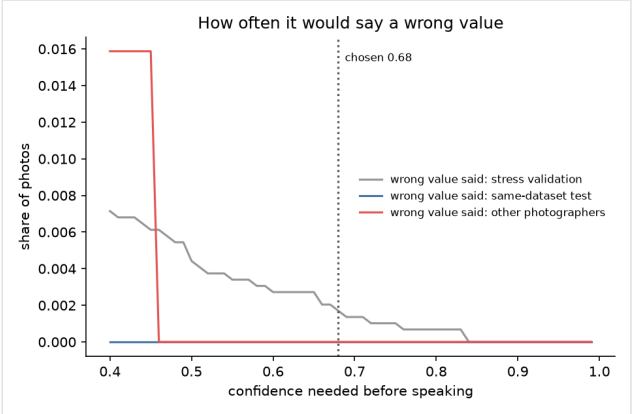


Figure 2. Share of photos on which the deployed model would say a wrong value, by threshold.

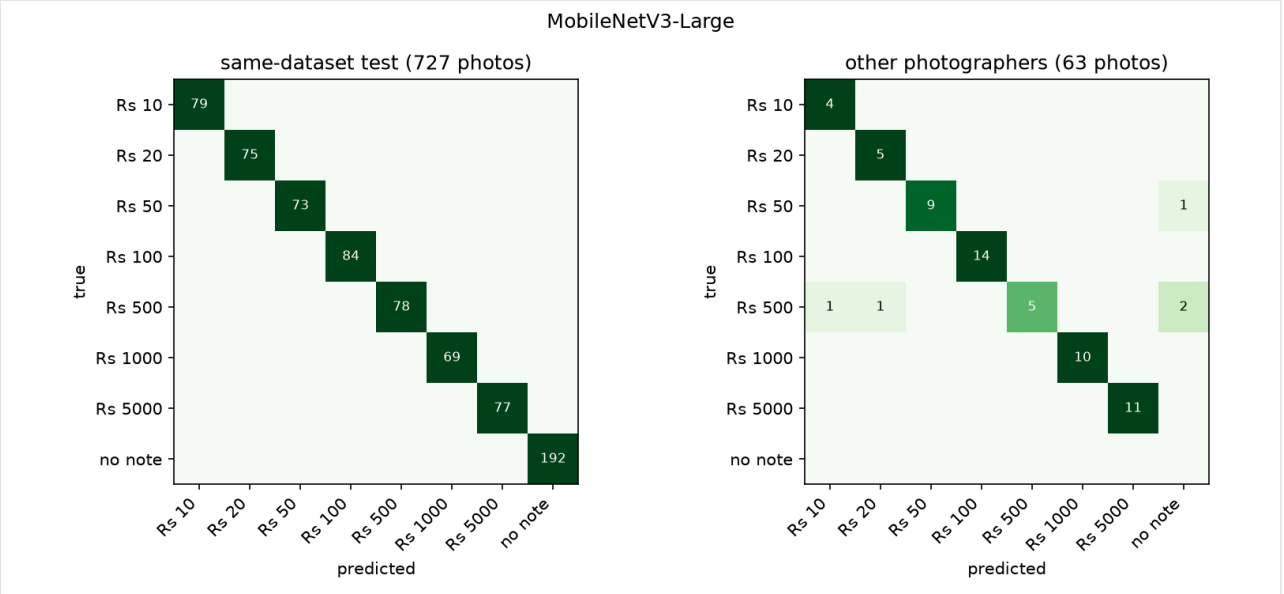


Figure 3. Confusion matrices of the deployed model on each test set.

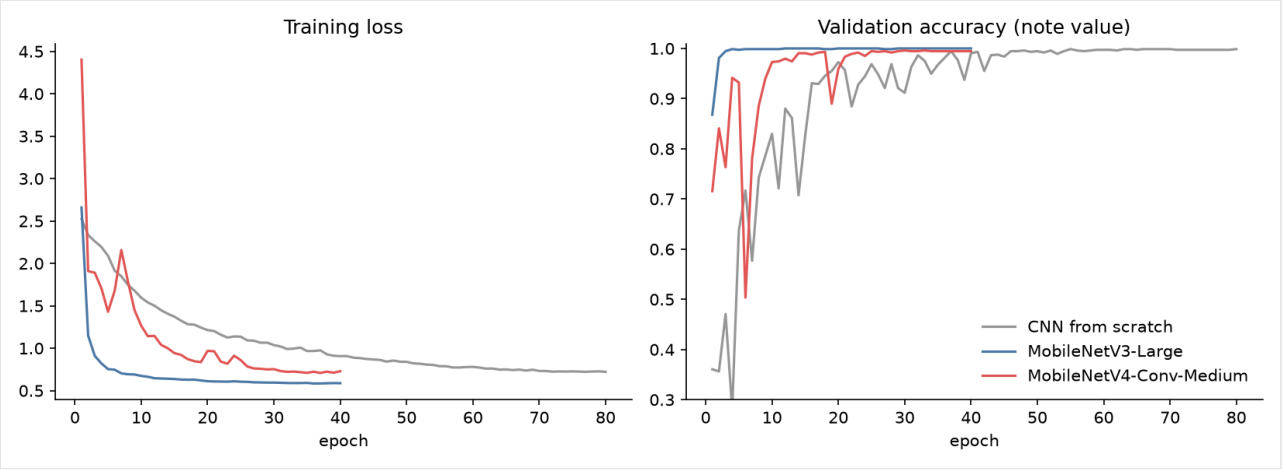


Figure 4. Training loss and validation accuracy.

5. Deployment

The deployed model is exported to ONNX (opset 17, 16.9 MB, softmax included) and TorchScript. The website runs it with ONNX Runtime Web on the phone, on the live rear camera, about four frames a second; the speech is pre-recorded with neural Urdu and English voices so it works even on phones without Urdu text-to-speech, and Android phones also vibrate a code (1–4 short pulses for Rs 10–100, 1–3 long for Rs 500–5,000). Results are announced in an ARIA live region so screen readers (TalkBack, VoiceOver) read them too. The repository holds code, splits, own photos, all weights with SHA-256 hashes in `model_card.json`, the executed notebook, and a GitHub Actions test.

6. Limitations

- The training photos show a few physical notes per value on similar indoor backgrounds; very worn, torn or taped notes are under-represented.
- It knows the 2005–2008 designs; the State Bank of Pakistan's announced new series will need new photos and retraining.

- One note at a time; it is not a counterfeit detector.
- The external test has only 63 photos, so its uncertainty is large (one photo ≈ 1.6 points).

7. Conclusion

A phone-sized network reads Pakistani banknotes aloud in Urdu, offline, in any browser. The headline in-dataset accuracy is close to perfect, but the honest numbers come from photos the dataset never saw, and a speaking rule tuned for safety matters as much as raw accuracy. Next: photos of worn notes, the new note series, counting several notes in one frame, and a field test with blind users.

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This report was drafted with the help of an AI assistant (Claude) from the project's code and results, as the course requires, and checked against them.